

Amazon Kinesis Interview Questions and Answers

Interview-style questions covering Kinesis Data Streams, Kinesis Data Firehose, shards, partition keys, consumer patterns, and how Kinesis compares to other streaming platforms.

Q1. What is Amazon Kinesis?

Amazon Kinesis is a family of managed services for collecting, processing, and analyzing real-time streaming data at scale. It includes Kinesis Data Streams, Kinesis Data Firehose, and Kinesis Data Analytics, each serving a different part of a streaming pipeline.

Q2. What is a shard in Kinesis Data Streams?

A shard is the base throughput unit of a Kinesis data stream. Each shard supports up to 1 MB per second of write throughput and up to 2 MB per second of read throughput (standard consumers), and up to 1,000 PUT records per second. A stream's total capacity depends on how many shards it has.

Q3. What is a partition key used for?

A partition key is supplied with each record and is hashed to determine which shard the record is written to. Choosing a partition key with high cardinality helps distribute records evenly across shards and avoids hot shards.

Q4. What is the difference between Kinesis Data Streams and Kinesis Data Firehose?

Data Streams gives you a low-latency stream that custom consumer applications can read from in near real time, with data retained for a configurable window. Firehose is a fully managed delivery service that automatically loads streaming data into destinations like S3, Redshift, or OpenSearch, without requiring you to write consumer code.

Q5. What is enhanced fan-out?

Enhanced fan-out gives each registered consumer a dedicated 2 MB per second read throughput per shard, delivered via a push model, rather than consumers sharing the shard's standard read throughput. This helps when multiple consumer applications need to read the same stream independently and quickly.

Q6. How does Kinesis handle data retention?

By default, records in a Kinesis data stream are retained for 24 hours, but retention can be extended up to 365 days for an additional cost. This allows consumers to reprocess or replay data within that retention window.

Q7. What is the Kinesis Client Library (KCL) used for?

KCL is a library that simplifies building consumer applications for Kinesis Data Streams. It manages load balancing across shards, checkpointing progress, and responding to shard splits or merges, so the developer can focus on record-processing logic.

Q8. How does Kinesis compare to Apache Kafka?

Both are used for real-time event streaming, but Kinesis is a fully managed AWS service with pricing based on shards and requests, while Kafka is typically self-managed or run via a managed offering like MSK, giving more control over configuration, retention, and partitioning at the cost of more operational responsibility.